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B. E. (Fifth Semester) Examination, April-May 2016

(New Scheme)

(Mechanical Engg. Branch)

OPERATIONS RESEARCH

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt for 20 marks from each question. Chart for normal neviate is permitted.*

1. (a) What are the advantages of Linear Programming Methods? 2

(b) Graphically find the minimum value : 9

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$$z = -x_1 + 2x_2$$

Subject to :

$$-x_1 + 3x_2 \le 10$$

$$x_1 + x_2 \le 6$$

$$x_1 - x_2 \le 2$$

$$x_1, x_2 \ge 0$$

(c) Solve by Simplex method : 9

Maximize $z = 4x_1 + 3x_2 + 6x_3$

Subject to

$$2x_1 + 3x_2 + 2x_3 \le 440$$

$$4x_1 + 3x_3 \le 470$$

$$2x_1 + 5x_2 \le 430$$

$$x_1, x_2, x_3 \ge 0$$

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(d) Solve the following LPP by Big-M-Method : 9

Minimize $z = 12x_1 + 20x_2$

Subject to

$$6x_1 + 8x_2 \geq 100$$

$$7x_1 + 12x_2 \geq 120$$

$$x_1, x_2 \geq 0$$

2. (a) Determine an initial basic feasible solution to the following transportation problem using North-West Corner Rule : 6

	To					Available
	3	4	6	8	9	20
	2	10	1	5	8	30
from	7	11	20	40	3	15
	2	1	9	14	16	13
Demand	40	6	8	18	6	

(b) Consider the following unbalanced transportation problem; if penalty cost for destination 1, 2 & 3 are 5, 3 & 2 respectively. 14

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To

Destination → / source ↓	1	2	3	Supply
1	5	1	7	10
from 2	6	4	6	80
3	3	2	5	15
Demand	75	20	50	

Find the optimal solution.

(c) Solve the following arrangement problem : 14

1	11	17	8	16	20
2	9	7	12	6	15
3	13	16	15	12	16
4	21	24	17	28	26
5	14	10	12	11	13

3. (a) A self-service store employs one cashier at its counter. Nine customers arrives on an average every five minutes. While the cashier can serve ten customers in 5 minutes. Assuming Poisson distribution for arrival rate and exponential distribution for service time, find :

(i) Average number of customers in the system

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(ii) Average time a customers waits before being served. 4

(b) A Project schedule has the following characteristics : 16

Activity	Time (Weeks)
1-2	4
1-3	1
2-4	1
3-4	1
3-5	6
4-9	5
5-6	4
5-7	8
6-8	1
7-8	2
8-10	5
9-10	7

(i) Construct the Network. 8

(ii) Compute E and L for each event. 5

(iii) Find the critical path. 3

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(c) Consider the network shown in figure. The three time estimates for the activities are given along the arrows :

(i) Determine the critical path. 9

(ii) Calculate standard devation for all i-j on critical path. 3

(iii) Determine the probability that the project will be completed in 20 days. 4

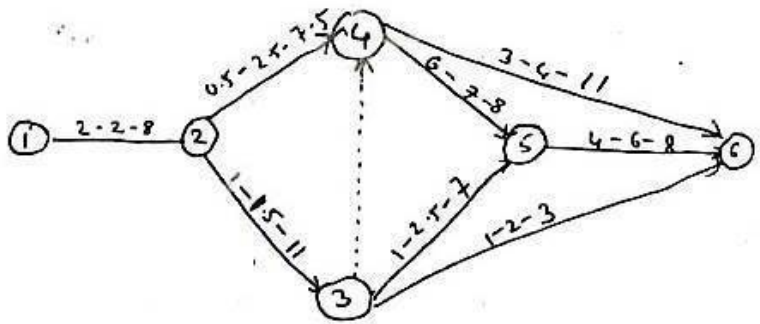

fig.

4. (a) What do you mean by two-person zero-sum game? What are its characteristics? 4

(b) Solve the following game by using principle of dominance : 16

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Player B

Player A	I	II	III	IV	V	VI
1	4	2	0	2	1	1
2	4	3	1	3	2	2
3	4	3	7	–5	1	2
4	4	3	4	–1	2	2
5	4	3	3	–2	2	2

(c) Two persons X and Y works on a two station assembly line. The distribution of activity times at their stations are tabulated below : 16

(i) Simulate operation of the line for eight items.

(ii) Assuming Y must wait unti X completes the first item before starting work, will he have to wait to process any of the other seven items? What is the average waiting time of items for Y?

Use the following Random numbers :

For X : 83 70 06 12 59 46 54 & 04

& For Y : 51 99 84 81 15 36 12 & 54

Given :

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Time in seconds (Sec)	Time frequency For (X)	Time frequency For (Y)
10	4	2
20	7	3
30	10	6
40	15	8
50	35	12
60	18	9
70	8	7
80	3	3

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